Scrum-volutional Neural Networks: Revolutionizing Agile Development with JIRA-Net

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Abstract

In the field of Artificial Intelligence, it is well understood that everything is made better with AI, or at least everything is more *marketable*. Therefore, in this paper, we take the beloved bug-tracking and sprint management tool JIRA and show its ability to be a neural network inference accelerator (or decelerator as the results may show).

Whilst Atlassian has yet to fully embrace JIRA as a deployment platform for Neural Networks, we believe that deploying AI features that no-one asked for is quickly becoming industry standard. With our novel convolutional neural network 'JIRA-Net', scrum masters can harness the power of A.I. to automate their most tedious of tasks, slow down their corporate on-site installation, and secure their future job prospects by being the only people who know how to turn this crap off.

With JIRA-Net, we detail how we achieve whelming 71% accurate digit recognition results in our scrum workflow and how you can build your own neural network inside the workplace's premier productivity tool.

We discuss the difficulties and decisions to be made when training a network in such a resource-constrained environment and ruin several of our peer-reviewers' days.

1 Introduction

> I toast, therefore I am
>
> *Talkie Toaster, Red Dwarf*

Some might say that Artificial Intelligence has gone too far, what with the stealing of artist's work [7], the environmental cost[11], the loss of jobs [8] and the prospect of reaching the Singularity [10]. Well, I think it hasn't gone far enough! Red Dwarf (1988) [2] gives us a glimpse of a possible luxurious future where we can engage in light conversation with our household objects, even low-compute toasting devices. In a world where we are becoming increasingly lonely [9], maybe some artificial connection is greater than no connection at all... (we request the reader pause meaningfully before reading on)

And so, whilst I sat waiting, yet again, for a JIRA ticket to submit through my corporation's various levels of indirection, bureaucracy, physical hand-overs, micro-outages and secret handshakes, I wondered — what if a toaster could submit this for me?

2 JIRA-Net

> The only thing we have to fear is fear itself (And JIRA)
>
> *Franklin D. Roosevelt, and also me.*

To execute a neural network, the most basic requirement we need is a system that is turing-complete. Thankfully, JIRA's automation tools [12] offer us enough convenient[1] facilities for both computation and memory storage.

However, there are limitations to this approach. JIRA lookup tables are limited to 200 items, are strictly one-dimensional and have inflexible formats for key values. While our original aim was to process the seminal LeNet network [1] on the platform, we instead had to build a bespoke model to fit into the constraints of the system. One that required no tensor be larger than 200 items.

Because of this, we begin on the back foot as the largest image we can consume is 1x14x14 (1/4 of the size of a regular MNIST image).

Dimensionality continually recurs as a pain point in designing a neural network fit to call itself agile. Particularly channel sizes pose a large problem - as increasing an activation's channel size proportionally increases the associated weight, resulting in a delicate balancing act. To avoid this, we took some inspiration from ResNet[3] where residual units sample different granularities before decreasing the activation width and height.

The full network architecture is given in Table 1 and Figure 1

2.1 Looking ahead to JIRA-Net v2

> The future is much like the present, only longer.
>
> *Dan Quisenberry*

In the second block, we use 1x3 and 3x1 convolutions instead of a 3x3 convolution to avoid increasing the overall parameter size. Looking back on this decision, it seems unnecessary as the parameters are small enough to not be much of an issue. We leave this and other fine-tuning decisions to future researchers to determine, as we are kind, benevolent and also, have already spent

[1] In strictly relative terms

more time than any sane person should on creating a JIRA-powered neural network. When this paper is published, we desire to never hear about it ever again.

However the wider topic of processing neural networks within specific memory limitations is actually an interesting one that may merit some actual research. Perhaps exchanging space preserved for biases with more weights, probing questionable decisions like the two identical Linear Layers at the end, experiment with using 3x3s2 poolings rather than 2x2s1 poolings, throw in a bunch of ReLUs for the craic, etc.

We believe that with enough effort, an accuracy score that is competitive with LeNet is genuinely possible. Whether it is worth spending all that effort on the next version of JIRA-Net, is more questionable.

2.2 Exact Architecture Details

Stop putting quotes in your papers, its unprofessional

Unnamed Advisor

Layer	**Kernel Options**	**Output**	**Parameters**
Network Input	Resize (Bilinear Interpolation)	1x14x14	—
Convolution '1A'	1x1	1x14x14	10
Convolution '1B'	3x3 pad 1	1x14x14	26
Convolution '1C'	5x5 pad 2	1x14x14	50
Eltwise		1x7x7	—
Maxpool	2x2	1x7x7	—
Convolution '2A'	1x1	2x7x7	4
Convolution '2B'	1x3 pad 0x1	1x7x7	4
Convolution '2C'	3x1 pad 1x0	1x7x7	4
Concat		4x7x7	—
Maxpool	2x2	4x3x3	—
Convolution '3A'	1x1	4x3x3	74
Convolution '3B'	3x3 pad 1	2x3x3	20
Concat		6x3x3	—
Convolution '4'	1x1	5x3x3	35
Convolution '5'	3x3	4x1x1	184
Linear Layer '6'		1x19	95
Linear Layer '7'		1x10	200
Linear Layer '8'		1x10	110

Table 1: JIRA-Net Architecture

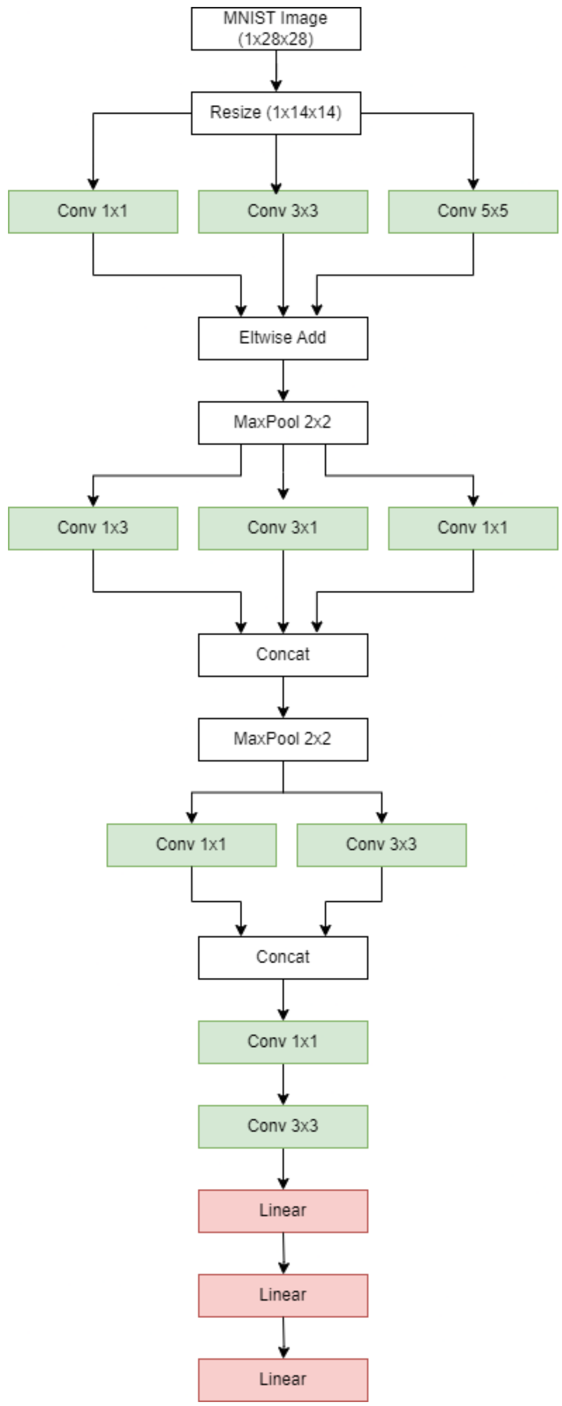

Figure 1: JIRA-Net, in all its glory

3 Deployment

If you can't solve a problem,
make it bigger

Anon

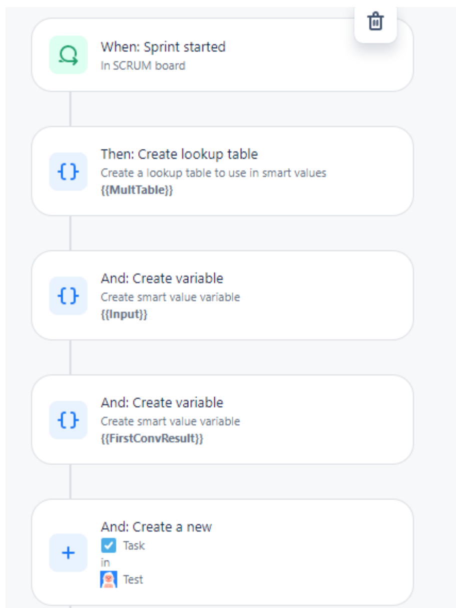

Figure 2: A single Neuron's progession through JIRA

JIRA Automation provides a drag-and-drop programming interface for creating your own sequence of items. At a high level, the implementation of our neural inference is simple:

3.1 Initialization

JIRA offer a wide selection of events that can be monitored for triggering automation workflows. We chose to run our neural network whenever we started our sprint, but you could have it trigger when an issue is assigned to you, at a scheduled interval, when you perform a release or even if your team breached your service level agreement (SLA) with its customers (Perhaps in the future you could automate an LLM to send an apology).

3.2 Tensor Storage

As mentioned in Section **??**, memory limitations are a big issue in JIRA. We are able to create a lookup table, but with only 200 values.

Plus, the lookup table accessors are numeric-only[2], so instead of a nice coordinate system to access pixels (e.g. "x:y" or "x_y"), we need to zero-pad our values (But, they aren't processed as strings, so if you use 0001 for the second pixel in the first row, you need to refer to it as 1).

In the future, we hope there is a way to import models into the JIRA Automation system so that we do not have to insert every single parameter by hand.

3.3 Computation

Thankfully, JIRA comes with math expression support. So we were able to perform the NN operations as follows (math for a single pixel only):

Convolution/Linear Layer

```
{{#=}}({{input.get(0)}} * {{weights.get(0)}}) + {{bias.get(0) }}{{/}}
```

ReLU

```
{{#=}} max({{input.get(0)}}, 0){{/}}
```

Maxpool

```
{{#=}} max({{input.get(0)}}, {{input.get(1)}, {{input.get(10)}, {{input.get(11)}}){{/}}
```

Eltwise

```
{{#=}} {{input.get(0)}} + {{input.get(11)}}{{/}}
```

Concat

Because JIRA does not have traditional Tensor structures and such, Concats can actually be optimized away. This is because you are effectively writing the network out by hand, neuron by neuron, so you can just address the correct pixel.

3.4 I/O

At present, our solution is sub-optimal from a user point of view, as you need to manual type the values into a lookup table for processing.

However, our output is much more sophisticated as we create a new bug report in our team's backlog with the title "Result: {THE_RESULT}" (Where 'THE_RESULT' is the NN digit prediction). An example of a single neuron's result from Convolution 1cA is shown in 3.

[2] Or at least, it always crashed whenever we tried anything else

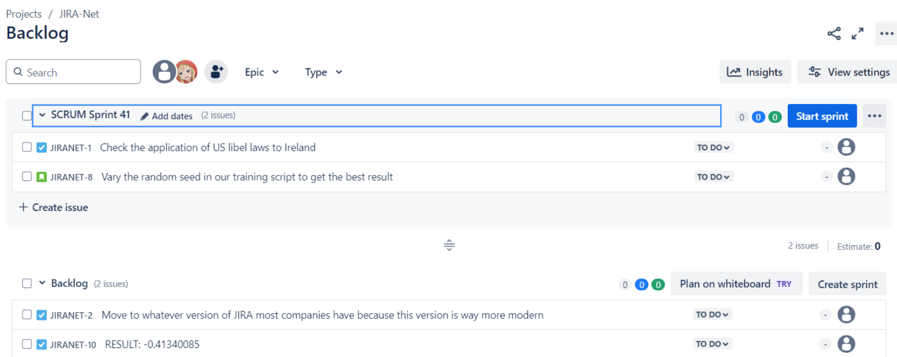

Figure 3: When the neural network is finished, it creates a ticket with your result

We encourage researchers who wish to build on this work to utilize the JIRA API [13], which allows querying for user avatars. This way all a user would have to do is change their avatar, begin a sprint and they will be informed of their result.[3]

4 Results

Just do your best

My mum

The results below speak for themselves (also, I'm tired).

[3]This may not be possible

4.1 Accuracy

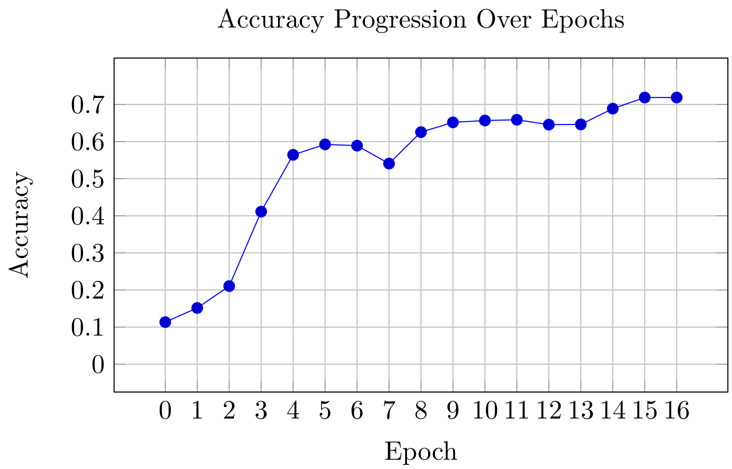

Figure 4: When it works [1], The network learns very fast before converging after a few number of epochs[2].

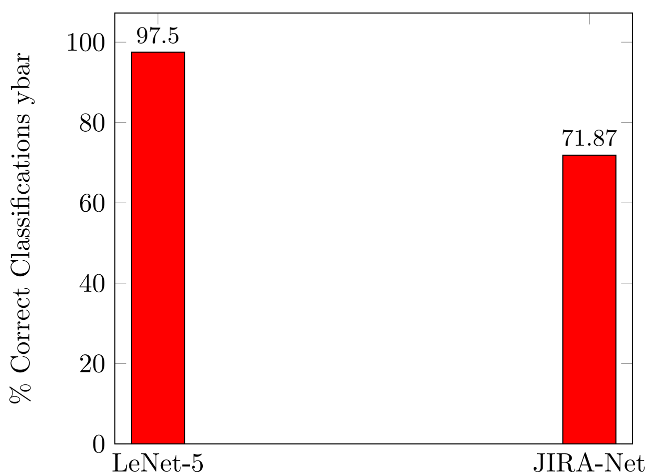

Figure 5: It's surprisingly not too bad considering I don't have much idea what I'm doing and my first step is to throw away half the image.

[3]100% of the time, 60% of the time

[3]Is that interesting? I don't know, but people often include these training diagrams and who am I to question it?

4.2 Performance

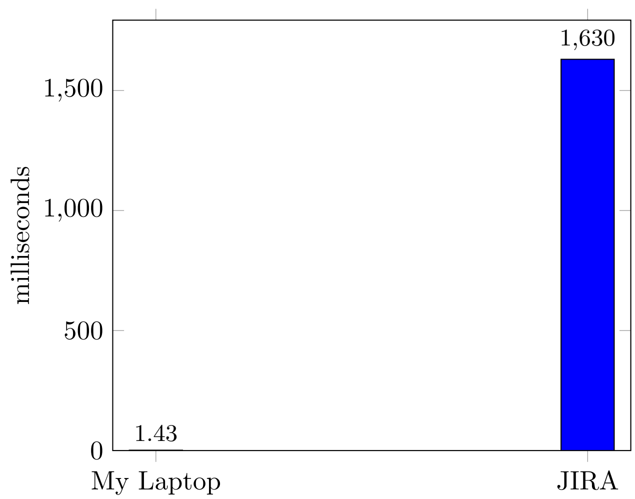

Figure 6: Performance Comparison: The first convolution of JIRA-Net on my laptop, versus a single pixel calculation of a convolution using JIRA as an inference engine

5 Conclusion

> You can't have a good idea if you don't have a bad idea first
>
> *Albert Einstein*

In conclusion, I think if you skim this paper and gloss over our results, you should still understand that JIRA-Net's existence shows the vast potential for JIRA to be a revolutionary platform for new AI models. It may indeed by slow and worse than the state of the art, but with further network training and hopefully Atlassian's enthusiastic embrace of this business segment, JIRA can be **the** next big thing in corporate bureaucracy.

We recommend that you, the reader, should implement a system like this for your own corporation, but blame another colleague.

5.1 Appendix: Disclaimer

We must admit that while we have proven JIRA-Net's accuracy offline and JIRA's ability to be an inference engine, we have not actually executed the entire network on JIRA. We attribute this short-coming to the fact that I did not have an intern to implement the many-thousand step process, and I would rather retract this paper than do it myself [4].

We did not receive any funding for this research (Nor should we have).

5.2 Appendix: Code

If you wish to re-implement JIRA-Net, or customize it for your own synergistic hybrid xtreme programming needs, you can find it here: GitHub Gist

5.3 Appendix: See Also

If our paper has gone further than others, it is because we are standing on the shoulders of giants. We must mention the influence that the 2016 SIGBOVIK paper "Deep Spreadsheets with ExcelNet" [4], the 2017 SIGBOVIK submission "On The Turing Completeness of PowerPoint"[6] and the MS Paint IDE[5] have had on the industry for companies other than Microsoft to consider making their own software bloated enough that someone thought it novel enough to mis-use it for secondary purposes.

References

[1] Yann LeCun et al. "Gradient-Based Learning Applied to Document Recognition". In: *Proceedings of the IEEE*. Vol. 86. 11. 1998, pp. 2278–2324. URL: `http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.42.7665`.

[2] Rob Grant and Doug Naylor. 2009.

[3] Kaiming He et al. *Deep Residual Learning for Image Recognition*. 2015. arXiv: `1512.03385 [cs.CV]`. URL: `https://arxiv.org/abs/1512.03385`.

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[5] *MS Paint IDE*. 2017. URL: `https://ms-paint-i.de/about`.

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[7] Julia Bausenhardt. *How ai is stealing your art*. Nov. 2024. URL: `https://juliabausenhardt.com/how-ai-is-stealing-your-art/`.

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[4] Also my premium trial ran out

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